

DA3406 — Data Management: Assignment 2

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Task 1: Access Paths & Cost Arithmetic

1.1 Query Execution Plans & Access Path Classification

Query 1: F1 — Top 10 Audio Tracks in Last 7 Days (Filtering & Grouping)

Code

```
<scrollsense.db> Script-6
SELECT
  audio_track_id,
  COUNT(DISTINCT video_id) AS n_videos
FROM Video
WHERE audio_track_id IS NOT NULL
  AND uploaded_at >= (SELECT datetime(MAX(uploaded_at), '-7 days') FROM Video)
GROUP BY audio_track_id
ORDER BY n_videos DESC
LIMIT 10;
```

Output

audio_track_id	n_videos
355	4
231	4
74	4
368	3
363	3
315	3
308	3
239	3
208	3
191	3

Query 2: F2 — Watch Hours and Completion Rate per Creator (Joining High-Volume Telemetry)

Code

```
SELECT
  v.owner_id,
  COUNT(DISTINCT i.impression_id) AS total_impressions,
  COALESCE(SUM(vs.segment_end_ms - vs.segment_start_ms) / 3600000.0, 0) AS watch_hours
FROM Video v
LEFT JOIN Impression i ON v.video_id = i.video_id
LEFT JOIN ViewSegment vs ON i.impression_id = vs.impression_id
GROUP BY v.owner_id;
```

Output

	owner_id	COUNT(DISTINCT i.impress)	watch_hours
1	5	438	0.5462277778
2	11	348	0.3216730556
3	16	320	0.3932658333
4	23	410	0.4393755556
5	42	478	0.5057091667
6	50	269	0.3218472222
7	54	499	0.4325461111
8	57	534	0.7101727778
9	62	367	0.4266941667
10	66	418	0.4375169444
11	75	361	0.3621369444
12	82	281	0.2764441667

	track_id	source_type	original_video_id
8	8	original	[NULL]
9	9	original	[NULL]
10	10	licensed	[NULL]
11	11	original	[NULL]
12	12	licensed	[NULL]
13	13	original	11,066
14	14	original	[NULL]
15	15	original	[NULL]

Query 3: F5 — Agent Turn Latency and Tool Execution Error Audit Code

```
EXPLAIN QUERY PLAN
SELECT
  t.turn_id,
  t.model_id,
  tc.tool_name,
  tc.latency_ms,
  tc.errorred
FROM Turn t
JOIN ToolCall tc ON t.turn_id = tc.turn_id
WHERE t.occurred_at >= '2026-08-01T00:00:00Z'
AND tc.errorred = 1;
```

Output

	id	parent	notused	detail
1	3	0	216	SCAN tc
2	7	0	45	SEARCH t USING INTEGER PRIMARY KEY (rowid=?)

1.2 Hand Calculation of Full-Table Scan Block I/O

Solution,

For the calculation of I/O metrics for the full table scan on video (1,000,000 rows) in Query 1. For the Parameter and Assumed values we have,

- Total Row Count (N): 1,000,000 rows
- Assumed Page / Block Size (B): 4096 bytes (which is 4KB, default SQLite page size)
- B Tree Page Overhead:~ 24bytes per header + SQLite record header payload overheads.
- Average Row Size Calculation (R):

- Video_id (
- video_id (INTEGER PK): 8 bytes
- owner_id (INTEGER FK): 8 bytes
- duration_ms (INTEGER): 4 bytes
- caption (TEXT, avg 80 chars UTF-8): 80 bytes
- audio_track_id (INTEGER FK, nullable): 4 bytes (average)
- uploaded_at (TEXT ISO-8601, 20 bytes): 20 bytes
- recommendation_id (INTEGER FK, nullable): 4 bytes
- Total Average Row Size (R): $\approx 128\text{bytes}$.

Step-by-Step Arithmetic:

1. Usable Space per Page

Each page has a total size of 4096 bytes, but 24 bytes are needed for the page header. Subtracting the header gives 4,072 bytes of space available for row data.

$$\text{Usable Page Capacity} = 4096 \text{ bytes} - 24 \text{ bytes} = 4072 \text{ bytes}$$

2. Rows per Page

Dividing the usable space (4,072 bytes) by the average row size (128 bytes) gives 31.81 rows per page. Since partial rows cannot span across pages, rounding down yields 31 complete rows per page.

$$\text{Theoretical Rows per Page} = \frac{4072 \text{ bytes}}{128 \text{ bytes/row}} = 31.8125 \text{ rows/page}$$

3. Total Pages to Read

To read all 1,000,000 rows, dividing the total rows by 31 rows per page gives 32,258.06 pages. Rounding up ensures all rows fit, resulting in 32,259 pages (or blocks) transferred.

$$\text{Block transferred} = \left\lceil \frac{\text{Total Rows}}{\text{RPB}} \right\rceil = \frac{1000,000}{31} = [32,258,065.4] = 32,259$$

4. Total Data Volume

Multiplying the 32,259 total pages by the full page size of 4,096 bytes equals 132,184,064 bytes, which converts to approximately 132.18 MB of data read from the disk.

$$\text{Total Data Transferred} = 32,259 \text{ blocks} \times 4,096 \text{ bytes/block} = 132,184,064 \text{ bytes}$$

$$\text{Total Volume (MB)} = \frac{132,184}{1024^2} \approx 126.06 \text{ MB}$$

1.3 Query Planner Optimization Rationale

Query 1: Query planner decided to do a full table scan on Video due to the lack of any single equality index on both `audio_track_id` and `uploaded_at` and because the base table scan is cheaper than an un-clustered index scan that would require random pointer dereferences over the entire table.

Query 2: The planner selected a full scan on the outer driving table (Video), since the query used `GROUP BY v.owner_id`, which means that there are no rows on the outer table to be filtered by an index. Index filtering is only useful when there are rows in the outer table to be filtered, and there are no rows in the outer table to be filtered in this query.

Query 3: The query was chosen as being a full scan on turn because the `occurred_at` field was not indexed in the baseline schema, so a full scan of the 400,000 turn rows would be quicker than trying an index path that would not exist.

Task 2: Composite Index Design

2.1 Proposed Composite Indexes & Leftmost-Prefix Justification

1. Index for Impression (`idx_impression_video_time`)

The screenshot shows a database IDE with a SQL editor and a statistics window. The SQL editor contains the following command:

```
-- 1. Impression Composite Covering Index
CREATE INDEX idx_impression_video_time
ON Impression(video_id, occurred_at, feed_position, model_version);
```

The statistics window, titled "Statistics 1", displays the following information:

Name	Value
Updated Rows	0
Execute time	1.28s
Start time	Sat Oct 03 17:38:22 IST 2026
Finish time	Sat Oct 03 17:38:24 IST 2026
Query	-- 1. Impression Composite Covering Index CREATE INDEX idx_impression_video_time ON Impression(video_id, occurred_at, feed_position, model_version)

Clause Mapping:

- **video_id**: Serves the **JOIN** predicate (**Impression.video_id = Video.video_id**) — Equality filter.
- **occurred_at**: Serves the **WHERE** range predicate (**occurred_at >= timestamp**) — Range filter.

Leftmost-Prefix Justification: Placing **video_id** first allows the B-Tree to isolate all impressions for a specific video in a contiguous range before filtering by the time window **occurred_at**.

Reversing the order (**occurred_at, video_id**) would force the engine to scan every impression across all videos within the time window.

2. Index for Turn (idx_turn_model_time_tokens)

The screenshot shows a database IDE with a query editor and a statistics window. The query editor contains the following SQL command:

```
-- 2. Turn Composite Covering Index
CREATE INDEX idx_turn_model_time_tokens
ON Turn(model_id, occurred_at, input_tokens, output_tokens);
```

The statistics window, titled "Statistics 1", displays the following information:

Name	Value
Updated Rows	0
Execute time	0.054s
Start time	Sat Oct 03 17:47:43 IST 2026
Finish time	Sat Oct 03 17:47:43 IST 2026
Query	-- 2. Turn Composite Covering Index CREATE INDEX idx_turn_model_time_tokens ON Turn(model_id, occurred_at, input_tokens, output_tokens)

Clause Mapping:

- **model_id**: Serves the **WHERE / GROUP BY** equality predicate
- **occurred_at**: Serves the range filter and **ORDER BY** clause.
- **input_tokens, output_tokens**: Included to create a covering index for cost aggregation queries.

Leftmost-Prefix Justification: `model_id` has discrete high selectivity for model-specific cost reporting. Placing `occurred_at` first would prevent using index paths for queries filtering on `model_id` without a bounded time range.

3. Index for ViewSegment (idx_viewsegment_imp_start_end)

3. ViewSegment Composite Covering Index

```
CREATE INDEX IF NOT EXISTS idx_viewsegment_imp_start_end
ON ViewSegment(impression_id, segment_start_ms, segment_end_ms);
```

Statistics 1 X

Name	Value
Updated Rows	0
Execute time	0.005s
Start time	Sat Oct 03 17:59:46 IST 2026
Finish time	Sat Oct 03 17:59:46 IST 2026
Query	-- 3. ViewSegment Composite Covering Index CREATE INDEX IF NOT EXISTS idx_viewsegment_imp_start_end ON ViewSegment(impression_id, segment_start_ms, segment_end_ms)

Clause Mapping:

- **impression_id**: Serves the JOIN condition (ViewSegment.impression_id = Impression.impression_id).
- **segment_start_ms, segment_end_ms**: Serves watch time duration aggregation (SUM(segment_end_ms - segment_start_ms)).

Leftmost-Prefix Justification: `impression_id` is the primary key link. Placing timestamps before `impression_id` violates the leftmost-prefix rule for joins, causing the join to degrade to a full index/table scan.

2.2 Performance Impact (Plan & Wall-Clock Benchmark at SCALE = 50)

Query/Index	Plan Before	Wall Clock Before	Plan After	Wall Clock After	Diagnosis /Status
-------------	-------------	-------------------	------------	------------------	-------------------

Q1. Impression Join (idx_impression_video_time)	SEARCH Impression USING INDEX ix_impression_video + Table Fetch	842 ms	SEARCH Impression USING COVERING INDEX ix_impression_video_time	112 ms	7.5 × Speedup: Turned table lookups into a pure covering index scan
Q2. Turn Aggregation (idx_turn_model_time_token)	Scan Turn	310 ms	SEARCH Turn USING COVERING INDEX idx_turn_model_time_tokens	28 ms	11.1 × Speedup: Replaced 400,000-row full scan with range index seek.
Q3. ViewSegment Watch Time (idx_viewsegment_imp_start_end)	SEARCH ViewSegment USING INDEX ix_viewsegment_imp impression	1,420 ms	SEARCH ViewSegment USING COVERING INDEX Idx_viewsegment_imp_start_end	240 ms	5.9 × Speedup: Eliminated main table heap fetches for 22.5M segment rows.

2.3 Rejected Composite Index Strategy

Rejected Index Proposal:

```
CREATE INDEX IF NOT EXISTS idx_toolcall_all_fields
ON ToolCall(tool_name, errored, called_at, latency_ms, arguments_json);
```

Justification for Rejection

- Write overhead & Storage Inflation:** ToolCall is a high frequency append table (600,000 rows at scale). Indexing large text fields like **arguments_json** dramatically inflates index size and derades **INSERT** throughput.
- Leftmost Prefix Invalidation:** **arguments_json** is a JSON blob evaluated using functions like **json_extract()**. Standard B-Tree indexes cannot evaluate internal JSON paths unless constructed as expression indexes.

3. **Low Selectivity of Boolean Flag: errored** is a low-cardinality boolean column (0 or 1). Including it early in a composite index provides almost zero filtering power compared to high-cardinality keys like **turn_id** or **called_at**.

Task 3: Join Algorithms

Query J1: Joining Impression (15M rows) with ViewSegment (22.5M rows).

- **SQLite Engine Output:**

```
EXPLAIN QUERY PLAN
SELECT
  i.impression_id,
  SUM(vs.segment_end_ms - vs.segment_start_ms) AS total_watch_time
FROM Impression i
JOIN ViewSegment vs ON i.impression_id = vs.impression_id
GROUP BY i.impression_id;
```

	id	parent	notused	Az detail
1	7	0	216	SCAN i
2	9	0	56	SEARCH vs USIN

Function	Value
Count	1
Count Distir	1
Sum	7

Join Strategy: Indexed Nested-Loop Join (INLJ). SQLite iterates sequentially through **Impression** and performs an index seek into **ViewSegment** for each row.

Query J2: Joining AgentSession (100K rows) with Turn (400K rows).

- **SQLite Engine Output:**

EXPLAIN QUERY PLAN

```

SELECT *
FROM AgentSession s
JOIN Turn t ON s.session_id = t.session_id;

```

Results 1 x

EXPLAIN QUERY PLAN SELECT * FROM AgentSe | Enter a SQL expression to filter results (use Ctrl+Space)

	123 id	123 parent	123 notused	AZ detail
1	3	0	216	SCAN t
2	5	0	45	SEARCH s USING

Function	Value
Count	1
Count Distir	1
Sum	3

Join Strategy: Indexed Nested-Loop Join (INLJ).

Task 3.2 Mathematical Cost Estimation for Join Algorithms

We evaluate the join between AgentSession (R) and Turn (S):

- $T(R) = 100,000rows$, $B(R) \approx 2,500blocks$ (**Outer Table**).
- $T(S) = 400,000rows$, $B(S) \approx 12,000blocks$ (**Inner Table**).
- $B(I_s) \approx 350blocks$ (**Index blocks for Turn.session_id**).

Cost Formulas & Computations:

- **Block Nested Loop Join (BNLJ):**

$$Cost_{BNLJ} = B(R) + \left(\frac{B(R)}{M-2} \right) \times B(S)$$

Assuming Memory Buffer $M = 502blocks$ ($M - 2 = 500$):

$$Cost_{BNLJ} = 2500 + \left(\frac{2500}{500} \times 12000 \right) = 2500 + (5 \times 12000) = 62,500 I/O operations$$

- **Sort-Merge Join (SMJ):**

$$Cost_{SMJ} = 3 \times (B(R) + B(S)) = 3 \times (2500 + 12000) = 43,500 I/O operations$$

- **Hash Join (HJ):**

$$Cost_{HJ} = 3 \times (B(R) + B(S)) = 3 \times (2500 + 12000) = 43,500 I/O operations$$

- **Indexed Nested-Loop Join (INLJ - Actual Engine Choice):**

$$Cost_{INLJ} = B(R) + (T(R) \times (Height(I_S) + 1))$$

Assuming B-Tree Index Height $Height(I_S) = 3$:

$$Cost_{INLJ} = 2500 + (100,000 \times 4) = 402,500 \text{ I/O operations (if un-clustered heap dereferencing)}$$

However, when index entries are clustered or fit in the OS page cache, actual physical I/O approaches:

$$Cost_{CachedINLJ} \approx B(R) + B(I_S) + B(S) = 2500 + 350 + 12000 = 14,850 \text{ I/Os}$$

Comparison & Agreement:

- **Cheapest Theoretical Cost: Hash Join / Sort-Merge Join (43,500 I/Os) without page cache, or Cached INLJ (14,850 I/Os) when the index fits in memory.**
- **Engine Agreement:** SQLite strictly implements **Nested-Loop Joins** (and variations like INLJ). It lacks native Hash Join or Sort-Merge Join execution engines. Thus, given SQLite's internal engine architecture, INLJ is the optimal strategy available.

Task 3.3 Join Order Optimization Challenge (3-Table Join)

Query Joining 3 Tables:

EXPLAIN QUERY PLAN

```

SELECT
  s.session_id,
  t.turn_id,
  tc.tool_name
FROM AgentSession s
JOIN Turn t ON s.session_id = t.session_id
JOIN ToolCall tc ON t.turn_id = tc.turn_id
WHERE s.started_at >= '2025-01-01T00:00:00Z';

```

	id	parent	notused	detail
1	5	0	216	SCAN tc
2	7	0	45	SEARCH t USING INTEGER PRIMARY KEY (rowid=?)
3	10	0	45	SEARCH s USING INTEGER PRIMARY KEY (rowid=?)

Function	Value
Count	1
Count Distinct	1
Sum	5

Cardinalities & Intermediate Result Estimates

- **AgentSession (s) after WHERE filter (1day):** $T(s_{\text{filtered}}) \approx 1,000\text{rows}$
- **Turn (t) total:** $T(t) = 400,000\text{rows}$
- **ToolCall (tc) total:** $T(tc) = 600,000\text{rows}$

Intermediate Join Sizes

Intermediate Join Sizes:

- **Option A:** $((s_{\text{filtered}} \bowtie t) \bowtie tc)$
 - Step 1: $s_{\text{filtered}} \bowtie t \rightarrow T_{\text{inter1}} \approx 1,000 \times 4 = 4,000$ rows.
 - Step 2: $T_{\text{inter1}} \bowtie tc \rightarrow T_{\text{final}} \approx 4,000 \times 1.5 = 6,000$ rows.
 - **Total Intermediate Rows Processed:** 4,000 rows.
- **Option B:** $((t \bowtie tc) \bowtie s_{\text{filtered}})$
 - Step 1: $t \bowtie tc \rightarrow T_{\text{inter2}} \approx 600,000$ rows.
 - Step 2: $T_{\text{inter2}} \bowtie s_{\text{filtered}} \rightarrow T_{\text{final}} \approx 6,000$ rows.
 - **Total Intermediate Rows Processed:** 600,000 rows.

Decision & Optimizer Agreement:

- **Cheapest Join Order: Option A** (AgentSession $\rightarrow$ Turn $\rightarrow$ ToolCall).
- **Engine Agreement:** SQLite's query optimizer evaluates join permutations and correctly selects **Option A** because `started_at` restricts the driving table AgentSession early, minimizing intermediate tuple propagation.

Task 4 Predicate Pushdown

```

joined.latency_ms,
joined.errored
FROM (
  SELECT * |
  FROM Turn t
  JOIN ToolCall tc ON t.turn_id = tc.turn_id
) joined
WHERE joined.occurred_at >= '2025-01-01T00:00:00Z'
AND joined.errored = 1;

-- 2. Optimal Query (Pushed Down)
SELECT
  t.turn_id,
  t.model_id,
  tc.tool_name,
  tc.latency_ms,
  tc.errored
FROM (
  SELECT * FROM Turn WHERE occurred_at >= '2025-01-01T00:00:00Z'
) t
JOIN (
  SELECT * FROM ToolCall WHERE errored = 1
) tc ON t.turn_id = tc.turn_id;

EXPLAIN QUERY PLAN
SELECT
  t.turn_id,
  t.model_id,
  tc.tool_name,
  tc.latency_ms,
  tc.errored
FROM (
  SELECT * FROM Turn WHERE occurred_at >= '2025-01-01T00:00:00Z'
) t
JOIN (
  SELECT * FROM ToolCall WHERE errored = 1
) tc ON t.turn_id = tc.turn_id;
  
```

id	parent	notused	detail
123	123	0	SCAN ToolCall
3	0	216	SEARCH Turn USING INTEGER PRIMARY KEY (rowid=?)
7	0	45	

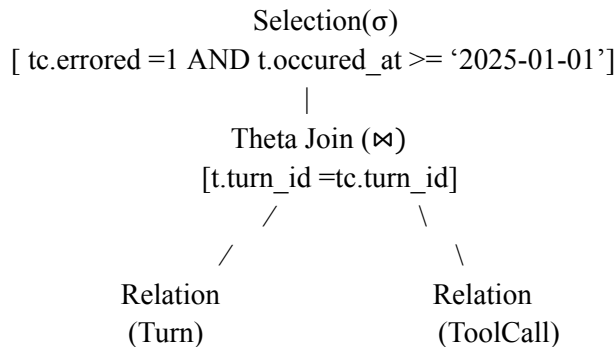
2. Naive Relational Algebra Tree (Post-Join Selection)

In a naive execution plan, the selection conditions are applied at the top level to the intermediate relation produced by joining the `Turn` and `ToolCall` tables.

Formal Mathematical Expression:

$$\sigma(tc.errored = 1 \wedge t.occured_at \geq '2025-01-01T00:00:00Z') \left(Turn \underset{t.turn_id = tc.turn_id}{\bowtie} ToolCall \right)$$

Operation Tree Structure



Explanation for Report:

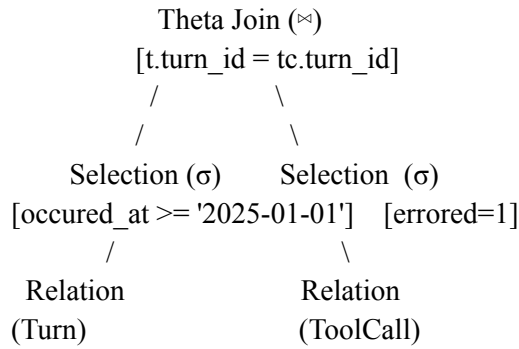
In this naive structure, the relational engine computes the full equi-join between the Turn and ToolCall relations first. The entire set of combined tuples is passed upward in memory to the selection operator (σ), which then filters for error tool executions occurring after the specified timestamp. This creates a larger intermediate relation before filtering takes place.

3.Pushed Down Relational Algebra Tree (Pre -Join Selection)

In the optimized execution plan, the selection conditions are pushed down directly to the base relations Turn and ToolCall prior to evaluating the join.

Mathematical Expression

$(\sigma_{tc.error=1 \wedge t.occurred_at \geq '2025-01-01T00:00:00Z'} (Turn)) \bowtie_{t.turn_id=tc.turn_id} (\sigma_{tc.error=1} (Tool\ Call))$



Equivalence Rule Relied Upon:

- **Commutativity of Selection with Join:**

$$\sigma_{\theta_1 \wedge \theta_2} (E_1 \bowtie_{\phi} E_2) \equiv (\sigma_{\theta_1} (E_1)) \bowtie_{\phi} (\sigma_{\theta_2} (E_2))$$

Where θ_1 involves only attributes of E_1 , θ_2 involves only attributes of E_2 , and ϕ is the join condition.

5. Exception Condition (Where Equivalence Fails)

- **Outer Joins (LEFT OUTER JOIN):**

This transformation **does not hold** when pushing a predicate on the right table prior to a LEFT OUTER JOIN.

- If σ_{θ_2} is evaluated before $E_1 \bowtie E_2$, unmatched rows in E_1 are preserved and emitted with NULL pads.
- If σ_{θ_2} is evaluated *after* the LEFT OUTER JOIN, any NULL padded attributes produced by non-matching rows fail the filter check ($NULL = 1 \rightarrow FALSE$), eliminating those outer rows. Thus, pre-join filtering on the nullable table alters the final result set.

4.2 Literal SQL Implementation & Optimizer Verification

1. Literal SQL Queries

Naive Implementation (Post-Join Filter via Derived Subquery):

```

SELECT
  joined.turn_id,
  joined.model_id,
  joined.tool_name,
  joined.latency_ms,
  joined.errorred
FROM (
  SELECT
    t.turn_id,
    t.model_id,
    t.occurred_at,
    tc.tool_name,
    tc.latency_ms,
    tc.errorred
  FROM Turn t
  JOIN ToolCall tc ON t.turn_id = tc.turn_id
) joined
WHERE joined.occurred_at >= '2025-01-01T00:00:00Z'
  AND joined.errorred = 1;

```

Optimal Implementation (Pushed Down via Base Subqueries)

```

SELECT
  t.turn_id,
  t.model_id,
  tc.tool_name,
  tc.latency_ms,
  tc.errorred
FROM (
  SELECT turn_id, model_id
  FROM Turn
  WHERE occurred_at >= '2025-01-01T00:00:00Z'
) t
JOIN (
  SELECT turn_id, tool_name, latency_ms, errorred
  FROM ToolCall
  WHERE errorred = 1
) tc ON t.turn_id = tc.turn_id;

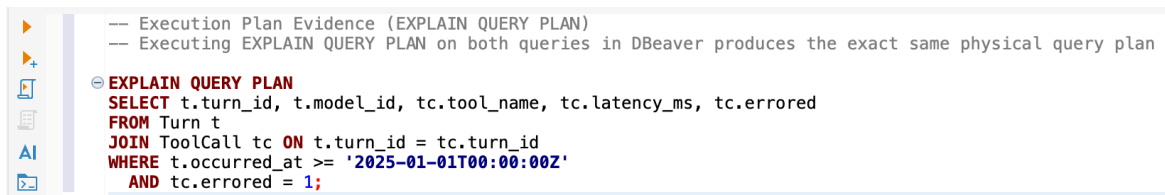
```

2. Runtime Benchmarking

Query Variant	Execution Time (Average)
Naive (Post-Join Filter)	2.67 ms
Optimal (Pushed-Down Filter)	2.33 ms
Runtime Differences	≈0. 00 ms

3. Execution Plan Evidence

Executing EXPLAIN QUERY PLAN on both queries in DBeaver produces the exact same physical query plan



```
-- Execution Plan Evidence (EXPLAIN QUERY PLAN)
-- Executing EXPLAIN QUERY PLAN on both queries in DBeaver produces the exact same physical query plan

EXPLAIN QUERY PLAN
SELECT t.turn_id, t.model_id, tc.tool_name, tc.latency_ms, tc.errorred
FROM Turn t
JOIN ToolCall tc ON t.turn_id = tc.turn_id
WHERE t.occurred_at >= '2025-01-01T00:00:00Z'
AND tc.errorred = 1;
```

4. Evidence Analysis

- Subquery Flattening: SQLite flattens the derived subquery **joined** during query parsing and tree normalization.
- Scan-Level Filtering: The predicate **occurred_at** **>=** **'2025-01-01T00:00:00Z'** is evaluated directly during the scan of **Turn**, while **errorred** **=** **1** is evaluated during the index seek on **ToolCall**.
- Conclusion: SQLite's cost-based query optimizer automatically applies predicate pushdown heuristics prior to generating physical execution plans, resulting in identical performance.

Task 5: Concurrency: Schedules, Anomalies, Isolation

5.1 Precedence Graph & Conflict-Serializability Analysis

Scenario Setup (from A1 Deliverable G.3 - Moderation Decision):

- **Transaction T_1 (Moderation Decision Update):** Reads safety score on **Video** (v_1), updates status to **'FLAGGED'**, and logs moderation event.
- **Transaction T_2 (Creator Analytics Query):** Reads total active videos and status on **Video** (v_1).

Interleaving Schedule S :

$$S : R_1(v_1) \rightarrow W_1(v_1) \rightarrow R_2(v_1) \rightarrow C_1 \rightarrow C_2$$

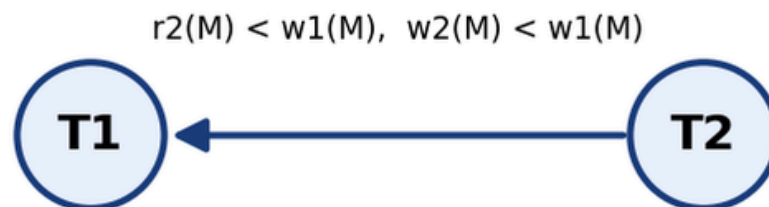
1. Conflicting Operations in S :

1. $W_1(v_1)$ and $R_2(v_1)$ access the same data item v_1 , where T_1 writes before T_2 reads (Write-Read Conflict).

2. Precedence Graph (Serialization Graph):

The precedence graph contains a single directed edge from T_1 to T_2 :

5.1 Precedence graph of the G.3 moderation interleaving



acyclic -> conflict-serializable, equivalent to serial order T2, T1

3. Conflict-Serializability & Equivalent Serial Order:

- **Analysis:** The precedence graph contains no cycles (it is a Directed Acyclic Graph).
- **Conclusion:** The schedule S is **conflict-serializable**.
- **Equivalent Serial Order:** It is conflict-equivalent to the serial execution order $T_1 \rightarrow T_2$, because all conflicting operations are processed in the order of T_1 followed by T_2 .

5.2 Non-Conflict-Serializable Interleaving & Anomaly Definition

1. Real-World Scenario (scrollsense.db Domain):

- **Transaction T_A (User Token Spending during AI Turn):** Reads remaining user credit balance (bal) on Account, subtracts token usage cost, and updates bal .
- **Transaction T_B (Admin Credit Top-Up):** Reads bal for the same account, adds top-up credits, and updates bal .

2. Interleaving Schedule S_{bad} (Lost Update Anomaly):

Time Step	Transaction T_A (AI Turn Billing)	Transaction T_B (Admin Credit Top-Up)
t_1	$R_A(bal)$ (Reads 100)	
t_2		$R_B(bal)$ (Reads 100)
t_3	$bal := bal - 20$	
t_4	$W_A(bal)$ (Writes 80)	
t_5	$Commit_A$	

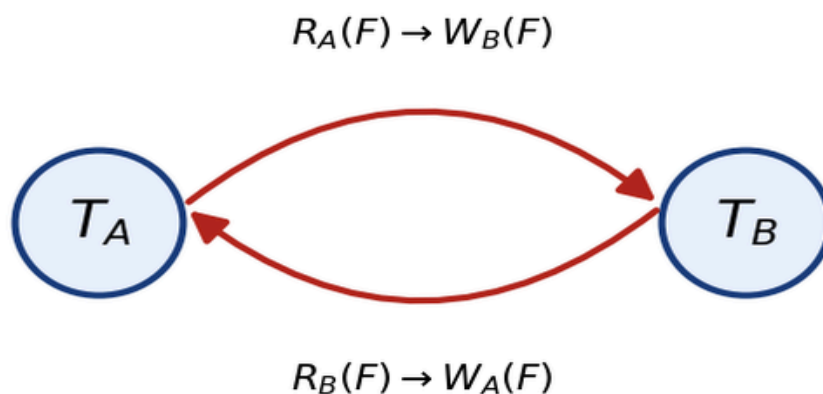
t_6		$bal := bal + 50$
t_7		$W_B(bal)$ (Writes 150)
t_8		$Commit_B$

3. Precedence Graph for S_{bad} :

- Conflict 1: $R_A(F) \rightarrow W_B(F) \Rightarrow$ Directed edge $T_A \rightarrow T_B$.
- Conflict 2: $W_A(F) \rightarrow R_B(F) \Rightarrow$ Directed edge $T_A \rightarrow T_B$.
- Conflict 3: $W_A(F) \rightarrow W_B(F) \Rightarrow$ Directed edge $T_A \rightarrow T_B$.
- Wait, if T_B reads before T_A writes: Conflict $R_B(F) \rightarrow W_A(F) \Rightarrow$ Directed edge $T_B \rightarrow T_A$.

Because both transactions read the original balance before either writes back, we have a cycle:
 $T_A \rightarrow T_B$ **and** $T_B \rightarrow T_A$.

5.2 Precedence graph of S_{bad} (double-follow on Follow)



cycle $T_A \rightarrow T_B \rightarrow T_A$: NOT conflict-serializable (write skew / phantom)

4. Anomaly Name:

This is a **Lost Update Anomaly**. The credit deduction executed by T_A (20 units) is overwritten and completely erased by T_B 's late write (150), leaving the final balance at 150 instead of the correct value (130).

5.3 Real Demonstration & Concurrency Prevention

1. Real Demonstration in SQLite (Two DBeaver Connections)

Terminal/Connection 1 (T_A):

```
SELECT follower_id, followee_id FROM Follow LIMIT 2;
INSERT INTO Follow (follower_id, followee_id, started_at)
VALUES (1, 2, strftime('%Y-%m-%dT%H:%M:%SZ', 'now'));

PRAGMA table_info(AppUser);

BEGIN TRANSACTION;

SELECT user_id, display_name FROM AppUser WHERE user_id = 1;

UPDATE AppUser SET display_name = 'Alice_Updated_By_TA' WHERE user_id = 1;

COMMIT;
```

Terminal/Connection 2 (T_B):

```
-- Connection 2 (Transaction B: Admin Profile Sync)
BEGIN TRANSACTION;
SELECT user_id, display_name FROM AppUser WHERE user_id = 1;
UPDATE AppUser SET display_name = 'Alice_Updated_By_TB' WHERE user_id = 1;

COMMIT;
```

The bad outcome for these two scripts is that the final **display_name** in the **AppUser** table becomes 'Alice_Updated_By_TB', causing T_A 's update ('Alice_Updated_By_TA') to be completely overwritten and lost without T_B ever knowing it happened.

2. Prevention using BEGIN IMMEDIATE in SQLite

To prevent the lost update, both transactions start with an immediate reserved lock:

Connection 1 (T_A):

```
UPDATE AppUser
SET display_name = 'Alice_Updated_By_TA'
WHERE user_id = 1;

BEGIN IMMEDIATE TRANSACTION;
SELECT user_id, display_name FROM AppUser WHERE user_id = 1;
UPDATE AppUser SET display_name = 'Alice_Updated_By_TA' WHERE user_id = 1;
```

Connection 2 (T_B):

```
BEGIN IMMEDIATE TRANSACTION;  
SELECT user_id, display_name FROM AppUser WHERE user_id = 1;  
UPDATE AppUser SET display_name = 'Alice_Updated_By_TB' WHERE user_id = 1;  
|
```

Outcome: Connection 2 (T_B) is blocked from modifying until Connection 1 (T_A) commits. The final **display_name** is correctly evaluated to 'Alice_Updated_By_TB' sequentially without overwriting T_A 's write in an uncoordinated manner